EAST WEST UNIVERSITY

Assignment 02 – Report

Course Title: Machine Learning

Course code: CSE475

Section: 03

# Title: Self-Supervised Learning for Eggplant Leaf Disease Classification: A BYOL and DINO Comparison

**Backbone Chosen: EfficientNet-B3**


| Submitted By | Submitted To |
|---|---|
| **Group – H**<br><br>**1. Asifa Akter Liya**<br>**ID: 2022-3-60-186**<br>**2. Riya Akter**<br>**ID: 2022-3-60-176**<br>**3. Benazir Meem**<br>**ID: 2022-3-60-169**<br>**4. Sadia Afrin**<br>**ID: 2022-2-60-088** | **Dr. Mohammad Rifat Ahmmad Rashid**<br>Assistant Professor,<br>Department of Computer Science and Engineering |


**Date of Submission: 23 April, 2026**

# Table of Contents


1. **Introduction** ..... 3
   - 1.1 **Motivation and Scope of Self-Supervised Learning (SSL)** ..... 3
   - 1.1 **Scope of BYOL and DINO** ..... 3
   - 1.2 **Reaserch Question** ..... 4
2. **Dataset and Preprocessing** ..... 5
   - 2.1 **Dataset Introduction** ..... 5
   - 2.2 **Reproducibility and Input Parameters** ..... 5
   - 2.3 **Exploratory Data Analysis (EDA)** ..... 5
   - 2.4 **Data Splitting Strategy** ..... 6
   - 2.5 **Dealing with Class Imbalance** ..... 6
   - 2.6 **Augmentation Strategy** ..... 6
   - 2.7 **Method-Specific Design Intent** ..... 7
3. **BYOL Implementation** ..... 7
   - 3.1 **Architecture** ..... 7
   - 3.2 **Training Configuration** ..... 8
   - 3.3 **Loss Curve Analysis** ..... 8
   - 3.4 **Convergence Analysis** ..... 9
4. **DINO Implementation** ..... 10
   - 4.1 **Architecture** ..... 10
   - 4.2 **Multi-Crop Strategy** ..... 11
   - 4.3 **Loss Curve Analysis** ..... 11
   - 4.4 **Attention Maps Visualization** ..... 12
5. **Evaluation and Results** ..... 14
   - 5.1 **Linear Probe Results** ..... 14
   - 5.2 **k-NN Classification Results** ..... 14
   - 5.3 **Overall Comparison with Baseline Methods** ..... 16
   - 5.4 **Per-Class Performance Analysis** ..... 16
6. **Discussion and Ablation** ..... 17
   - 6.1 **Ablation Results** ..... 17
   - 6.2 **Reflection** ..... 18
   - 6.3 **Limitations** ..... 18
7. **Conclusion and Future Work** ..... 19

**7.1** Conclusion ..... 19
7.2 Future Work ..... 20
**References** ..... 21

# 1. Introduction

## 1.1 Motivation and Scope of Self-Supervised Learning (SSL)

Supervised learning has shown great success in various computer vision applications; nevertheless, it requires the existence of large-scale labeled data. On the one hand, manually annotating such data is costly and inefficient. On the other hand, it can be biased owing to the subjectivity of humans involved. In practice, having good quality labeled data for training machine learning models can become a challenge.

The abundance of unlabeled data prompted researchers to consider another learning approach called Self-Supervised Learning (SSL). SSL learns generalizable feature representations without using human-labeled data by creating some pretext task or objective. This learning method allows us to improve data efficiency, scalability, and obtain transferable features, which can perform well on various applications, such as image classification, object detection, semantic segmentation, and even retrieval. Therefore, SSL is an emerging and promising research area in computer vision.

Various SSL models have been developed over time. In this report, the discussion is centered on two non-contrastive SSL frameworks that have been widely studied. These two models include Bootstrap Your Own Latent (BYOL) and Distillation with No Labels (DINO). The first SSL model, BYOL, is characterized by the use of online networks to learn the representations of targets produced by the target network. Importantly, BYOL does not use any negative examples, which are common in contrastive learning. Therefore, the training process is made more efficient while delivering excellent results. The second SSL model, DINO, is defined by the concept of self-distillation through the teacher–student network approach. It involves the generation of target representations by the teacher network and matching them using the student network. Importantly, it works efficiently in Vision Transformers since it can learn highly semantic visual information without supervision.

In terms of the scope of this paper, it is important to restrict our focus to the discussion of two models only, namely BYOL and DINO. It is done deliberately in order to allow us to conduct a comparative analysis of these two frameworks.

## 1.1 Scope of BYOL and DINO

The scope of this paper pertains to two major unsupervised SSL approaches used to learn representations from unlabeled images – Bootstrap Your Own Latent (BYOL) and Distillation with No Labels (DINO). The specific objective of this work under the given scope is to examine how each of these SSL approaches learns effective visual representations without using class labels, along with discussing the characteristics and behaviors of the approaches.

In terms of BYOL, the scope entails the approach's unique online-target network structure, whereby the online network is trained to predict representations generated by a target network that is updated at a slow pace. In particular, the online-target network architecture involves a momentum-based update rule used for updating the target network. Moreover, representation learning in BYOL is performed using the principle of predicting consistent representations of augmented views without utilizing any negative examples.

For DINO, the scope refers to the self-distillation paradigm, where the approach makes use of a teacher-student architecture. Here, the teacher network is updated through an exponential moving average of the student network, while the latter tries to mimic the predictions made by the former. The core components of the method that make it successful in learning visual representations include temperature scaling, output centering, as well as multi-view/multi-crop augmentation.

As far as the comparative scope is concerned, the comparison will focus on analyzing the quality of learned visual representations, task-specific performance, and stability during training. At the same time, other practical criteria such as sensitivity to hyperparameters, computation cost, and convergence will be examined to provide a comprehensive understanding of the differences between both approaches. All the experiments will take place in the same setup using the same dataset, backbone architecture, and evaluation metrics.

It is important to note that this study is limited to comparing the selected pair of algorithms rather than providing a review or comparison of self-supervised learning techniques in general. Moreover, the comparison will be limited to one single dataset and will not cover different types of domains and their respective challenges.

## 1.2 Reaserch Question

The current research stems from the requirement to analyze and compare contemporary self-supervised learning approaches for obtaining visual representations without relying on contrastive learning schemes. Specifically, the work explores differences in obtaining transferable features by means of Bootstrap Your Own Latent (BYOL) and Distillation with No Labels (DINO) algorithms when applied to unlabeled images.

To achieve these goals, the following research questions are formulated:

**Q1.** How effectively do BYOL and DINO learn meaningful visual representations from unlabeled image data?

**Q2.** Which method produces more transferable features for downstream classification tasks under the same training conditions?

**Q3.** How does the absence of negative samples in BYOL affect representation learning compared to DINO?

**Q4.** How does the teacher–student self-distillation mechanism in DINO influence feature quality?

**Q5.** What is the impact of momentum-based target network updates in BYOL on training stability?

**Q6.** How does multi-crop augmentation in DINO affect representation richness and semantic learning?

**Q7.** Which method converges faster during pretraining under identical experimental settings?

**Q8.** How do BYOL and DINO compare in terms of training stability and loss behavior over epochs?

**Q9.** What differences are observed in feature separability when evaluated using linear probing?

**Q10.** How do both methods perform under k-NN evaluation for representation quality assessment?

**Q11.** Which method is more computationally efficient in terms of training time and resource usage?

**Q12.** How sensitive are BYOL and DINO to hyperparameter changes such as learning rate and temperature scaling?

**Q13.** How do learned representations differ qualitatively in terms of attention patterns or feature clustering?

**Q14.** Which method generalizes better across unseen validation/test data?

**Q15.** What are the practical trade-offs between BYOL and DINO in terms of accuracy, stability, and efficiency?

# 2. Dataset and Preprocessing

In this section, we introduce our dataset, followed by a description of the dataset preprocessing techniques, exploratory data analysis (EDA), dataset splitting policy, and augmentation pipeline employed for BYOL and DINO self-supervised learning approaches.

## 2.1 Dataset Introduction

The chosen dataset for this work is the Eggplant Leaf Disease Classification Dataset with 2,991 samples of leaf images of eggplants classified by several disease classes. The dataset is organized in a class-wise directory structure and can be read by means of the ImageFolder utility.

Even though there are provided labels, they do not take place in the course of self-supervised pre-training and are solely applied to the evaluation procedures. Thus, the label information serves the purpose of downstream evaluation of representation learning quality through linear probing and validation.

## 2.2 Reproducibility and Input Parameters

For purposes of experiment reproducibility, random seed values were defined for the entire set of used modules (i.e., Python, Numpy, and PyTorch) in order to maintain consistency of dataset partitioning and training.

For the purpose of evaluation and linear probing, the size of the input images was set to 224 × 224 pixels. Yet, for self-supervised pre-training with SSL methods, the crops of different sizes were defined according to the adopted model.

## 2.3 Exploratory Data Analysis (EDA)

EDA was conducted to examine dataset features and to determine the preprocessing steps. These were some of the tasks carried out under EDA:

- Calculation of total samples, number of classes, and class distribution.
- Visualization of class imbalance by means of bar graphs.
- Evaluation of example images of each class.
- Checking the results obtained after augmenting the dataset through visualization of samples.

In this dataset, class imbalance and variations within a class were obvious.

## 2.4 Data Splitting Strategy

The dataset was divided using an **80/10/10 split**:

- **80% (2,392 images):** Self-supervised pretraining (unlabeled usage)
- **10% (299 images):** Linear probe / validation set
- **10% (300 images):** Final test set

All splits were carefully verified to ensure **no overlap between training, validation, and test sets**, maintaining strict dataset integrity.

The rationale behind this split is to maximize unlabeled data for SSL representation learning while preserving sufficient labeled samples for reliable downstream evaluation.

## 2.5 Dealing with Class Imbalance

Class imbalance is present in this dataset, meaning that some disease categories have many samples while other disease categories do not. To avoid any bias when training our model, we use WeightedRandomSampler in our data loading pipeline, especially for SSL training pipelines.

## 2.6 Augmentation Strategy

Data augmentation is a critical component of self-supervised learning, enabling models to learn invariant and robust representations.

### BYOL Augmentation Strategy

BYOL uses a **two-view consistency learning framework**, where two augmented versions of the same image are generated. The augmentation pipeline includes:

- RandomResizedCrop (224)
- RandomHorizontalFlip
- ColorJitter (0.4, 0.4, 0.4, 0.1)
- RandomGrayscale (p = 0.2)
- GaussianBlur

This strategy enforces **invariance learning by matching two strongly augmented views**, helping BYOL learn consistent representations across transformations.

### DINO Augmentation Strategy

DINO employs a **multi-crop augmentation strategy**, consisting of both global and local views:

- **Global crops (semantic views):**
  - RandomResizedCrop
  - RandomHorizontalFlip
  - ColorJitter
  - RandomGrayscale
  - GaussianBlur
  - Solarization
  - Normalization (channel-wise mean and standard deviation)
- **Local crops (fine-detail views):**
  - RandomResizedCrop (smaller scale)
  - RandomHorizontalFlip
  - ColorJitter
  - RandomGrayscale
  - GaussianBlur
  - Normalization

Channel-wise normalization is applied using standard ImageNet statistics to stabilize training and improve convergence.

This multi-view design enables DINO to capture both **global semantic structure and local discriminative details**, improving representation richness.

### 2.7 Method-Specific Design Intent

- **BYOL:** Learns invariant representations by matching two strong augmented views of the same image without using negative samples.
- **DINO:** Uses multi-crop self-distillation to learn both global semantics and fine-grained local patterns through teacher–student learning.

In general, the pre-processing pipeline is aimed at providing equal footing for both BYOL and DINO methods when working on the same datasets, as well as implementing method-specific augmentations. All the steps of the pre-processing pipeline were made based on the properties of the chosen datasets and insights from EDA.

## 3. BYOL Implementation

This section describes the implementation details of the Bootstrap Your Own Latent (BYOL) framework, including its architecture, training configuration, loss behavior, and convergence analysis.

### 3.1 Architecture

The BYOL model is built using **EfficientNet-B3** as the backbone network, where the final classification layer is removed (`nn.Identity`) to obtain high-dimensional feature embeddings (dimension: 1536).

The framework consists of two main components: an **online network** and a **target network**.

- **Online Network:**
  - `online_backbone`: EfficientNet-B3 feature extractor
  - `online_projector`: Multi-layer perceptron (MLP: 1536 → 4096 → 4096 → 256)
  - `predictor`: MLP (256 → 4096 → 4096 → 256) with Batch Normalization and ReLU activations
- **Target Network:**
  - `target_backbone`: Exponential Moving Average (EMA) copy of the online backbone
  - `target_projector`: MLP (1536 → 4096 → 4096 → 256)

The target network parameters are **not updated via backpropagation** (`requires_grad=False`). Instead, they are updated using EMA of the online network parameters. The BYOL learning objective is symmetric and defined as:

$$L = L(p1, t2) + L(p2, t1)$$

where p denotes online predictions and ttt denotes target projections from different augmented views.

## 3.2 Training Configuration

The BYOL model was trained using the following configuration:

- **Batch size:** 32
- **Number of epochs:** 30
- **Optimizer:** AdamW
- **Learning rate:** $3 \times 10^{-4}$
- **Weight decay:** $1 \times 10^{-6}$
- **Device:** CUDA-enabled GPU (when available)

The target network is updated using an EMA momentum schedule defined as:

$$\tau = 0.996 + \frac{epoch}{epochs - 1} \times (1 - 0.996)$$

This ensures that the momentum coefficient gradually increases from approximately 0.996 toward 1.0 during training. The target update rule is:

$$\theta_{target} \leftarrow \tau\, \theta_{target} + (1 - \tau)\, \theta_{online}$$

## 3.3 Loss Curve Analysis

The training loss behavior of BYOL is analyzed using the loss curve recorded over 30 epochs, as shown in **Figure 3.1**. The curve demonstrates a consistent downward trend, indicating effective optimization and progressive learning of meaningful representations from unlabeled data.

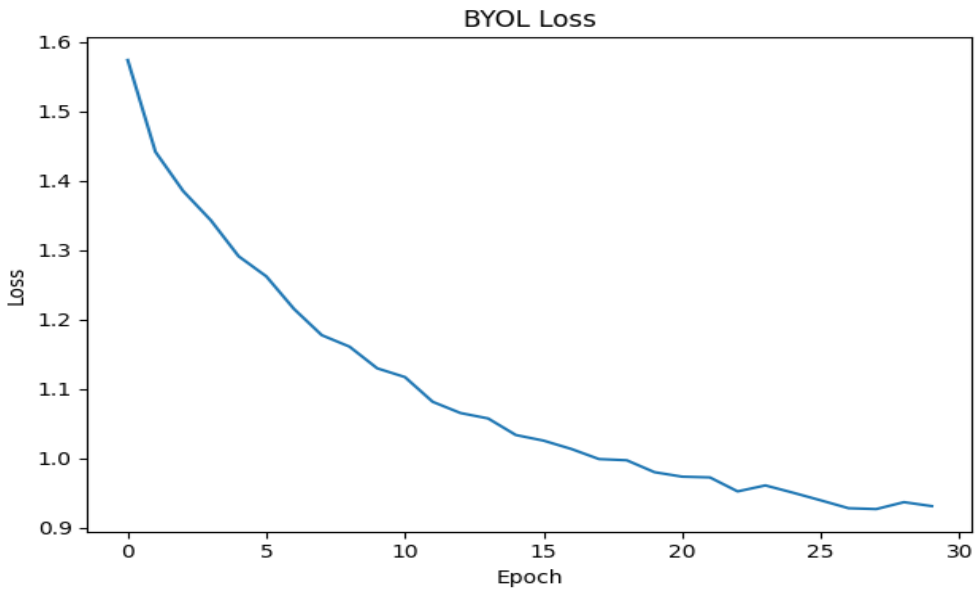


**Figure 3.1: BYOL training loss curve over 30 epochs**

Initially, the loss is relatively high, with a value of **1.5736 at epoch 1**, reflecting random or unrefined feature representations at the start of training. During the early training phase (approximately epochs 1–10), the loss decreases rapidly, reaching **1.1294 at epoch 10**, which suggests that the model quickly learns basic invariant features from augmented views.

In the mid-training phase (around epochs 10–20), the rate of improvement slows, with the loss reducing further to approximately **0.9798 at epoch 20**. This behavior indicates that the model transitions from coarse representation learning to finer feature refinement.

In the later stages of training, the loss begins to stabilize with minor fluctuations. The lowest loss region is observed around **epochs 27–28 (≈ 0.9278–0.9266)**, while the final recorded loss at epoch 30 is **0.9309**. These small oscillations in the final epochs are expected in self-supervised learning due to stochastic augmentations and exponential moving average (EMA) updates in the target network.

Overall, the loss curve in **Figure 3.1** confirms that BYOL achieves stable convergence, with most significant improvements occurring in the early epochs followed by gradual refinement and eventual plateauing of the training loss.

## 3.4 Convergence Analysis

Convergence analysis of BYOL indicates well-behaved and stable training dynamics. The first notable boost is observed for the first 15-20 epochs, representing fast learning of representations in the initial stage.

Following the previous stage, the second stage begins, characterized by a slower training process and little improvements in performance. Small oscillations in the following epochs (small changes from epoch 23 and beyond), which can be considered normal, since such changes are typical for self-supervised methods.

Finally, the flat loss at approximately 0.93 in the end means that the model converges to an almost optimal point in 30 epochs (configuration: AdamW optimizer, momentum scheduling, EfficientNet-B3 backbone).

In conclusion, BYOL exhibits successful and stable convergence properties, verifying the effectiveness of learning representations without negative examples.

# 4. DINO Implementation

This section describes the implementation of the **DINO (Distillation with No Labels)** framework, including its architecture, multi-crop augmentation strategy, loss behavior, and attention-based feature visualization.

## 4.1 Architecture

The DINO model is built on **EfficientNet-B3** as the backbone network, used for both student and teacher branches. The teacher network is a **momentum-updated (EMA) version** of the student network and remains frozen during backpropagation.

A key component of the architecture is the **DINOHead**, a hierarchical 3-layer MLP designed to project backbone features into a high-dimensional representation space:

- Input dimension: **1536 (EfficientNet-B3 embedding)**
- Hidden dimensions: **2048 → 2048**
- Output dimension: **4096 (normalized probability space)**

The architecture follows:

- Linear(1536 → 2048) + BatchNorm + GELU
- Linear(2048 → 2048) + BatchNorm + GELU
- Linear(2048 → 4096) (projection layer)
- Final projection to **4096-dim output space** (orthogonally initialized for stability)

Key characteristics include:

- **Student model:** Fully trainable via backpropagation
- **Teacher model:** EMA copy of student (no gradients)
- **Output space:** 4096-dimensional normalized probability distribution (higher expressiveness than BYOL)
- **Stability design:** Orthogonal initialization + batch normalization improves convergence stability

## 4.2 Multi-Crop Strategy

DINO employs a **multi-crop augmentation strategy** to enforce consistency across multiple spatial scales and improve representation richness.

Each input image generates **8 views total**, consisting of:

### Global Crops (Teacher Input)

- 2 crops per image
- Resolution: **224 × 224**
- Scale range: **0.4 – 1.0**
- Purpose: Preserve semantic-level global structure

### Local Crops (Student Input)

- 6 crops per image
- Resolution: **96 × 96**
- Scale range: **0.05 – 0.4**
- Purpose: Capture fine-grained local patterns and spatial details

### Batch Structure

- Each batch of images produces multiple crop views
- Total views per image: **8 (2 global + 6 local)**
- WeightedRandomSampler is applied to handle class imbalance in the SSL pool

This multi-view design enables the model to learn both **global semantic consistency** and **local discriminative features**, which is a key advantage over BYOL's simpler two-view setup.

## 4.3 Loss Curve Analysis

The DINO model is trained over **100 epochs**, and the loss behavior shows distinct phases of convergence.

- **Epoch 1:** 8.3116
- **Epoch 20:** 7.7786
- **Epoch 30–100:** Stabilization around ~8.32

### Training Behavior

- **Early phase (Epoch 1–20):**
  The loss decreases from 8.31 to 7.78, indicating rapid initial representation learning.
- **Mid phase (Epoch 20–30):**
  Slight fluctuations are observed as the teacher network stabilizes under EMA updates and temperature scheduling.

- **Late phase (Epoch 30–100):**
  The loss stabilizes around **8.32**, indicating convergence of the student–teacher system.

### Key Observations

- DINO converges faster than BYOL, reaching stability around **epoch 30**
- The higher loss magnitude compared to BYOL is due to **cross-entropy-based distillation**, which operates on a different scale than contrastive loss
- EMA momentum scheduling (0.996 → 1.0) and weight decay scheduling contribute to stable convergence
- Minor fluctuations in later epochs are expected due to stochastic multi-crop augmentation

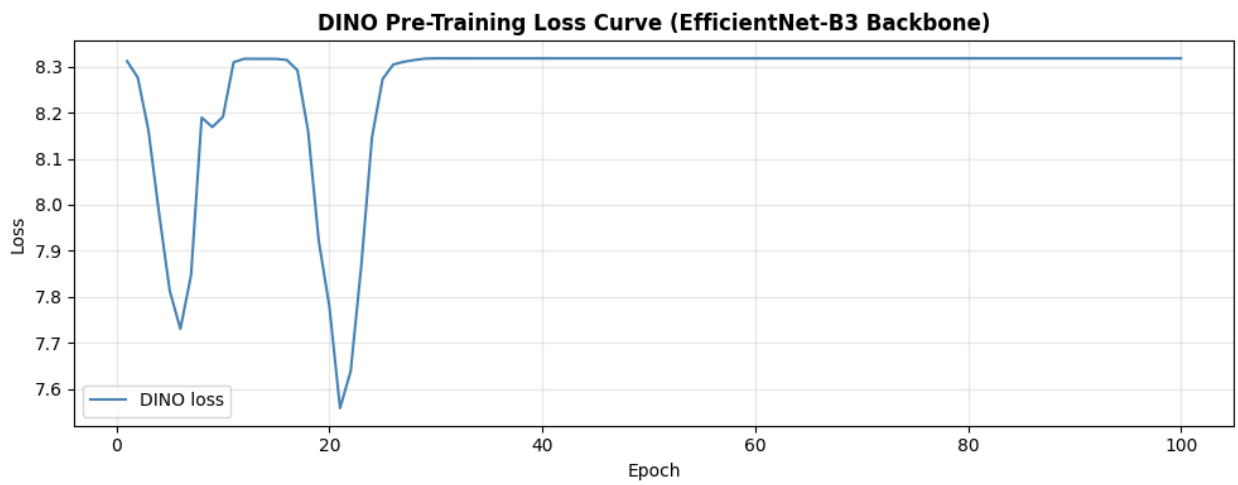


**Figure 4.1: DINO training loss curve over 100 epochs**

The training behavior of the DINO model over 100 epochs is illustrated in **Figure 4.1**. The loss initially decreases during the early epochs and then stabilizes, indicating convergence of the student–teacher network. Minor fluctuations are observed in later epochs due to stochastic augmentations and EMA-based updates.

## 4.4 Attention Maps Visualization

To interpret learned representations, **Grad-CAM-based attention maps** are generated using the last convolutional block of EfficientNet-B3.

### Methodology

- Forward hooks are applied to the final convolutional layer
- Feature maps of shape (C, H, W) are extracted
- Channel-wise averaging is applied to obtain spatial importance maps
- ReLU activation is used to retain only positive responses
- Maps are normalized and resized to **224 × 224** using bilinear interpolation
- Final heatmaps are overlaid on original images with 50% transparency

### Interpretation

The resulting attention maps highlight disease-relevant regions in eggplant leaves:

- **High activation regions (red):** Disease-affected leaf areas
- **Low activation regions (blue):** Background or irrelevant regions
- The model consistently focuses on biologically meaningful structures rather than noise

These visualizations confirm that DINO learns **semantically meaningful and spatially consistent representations**.

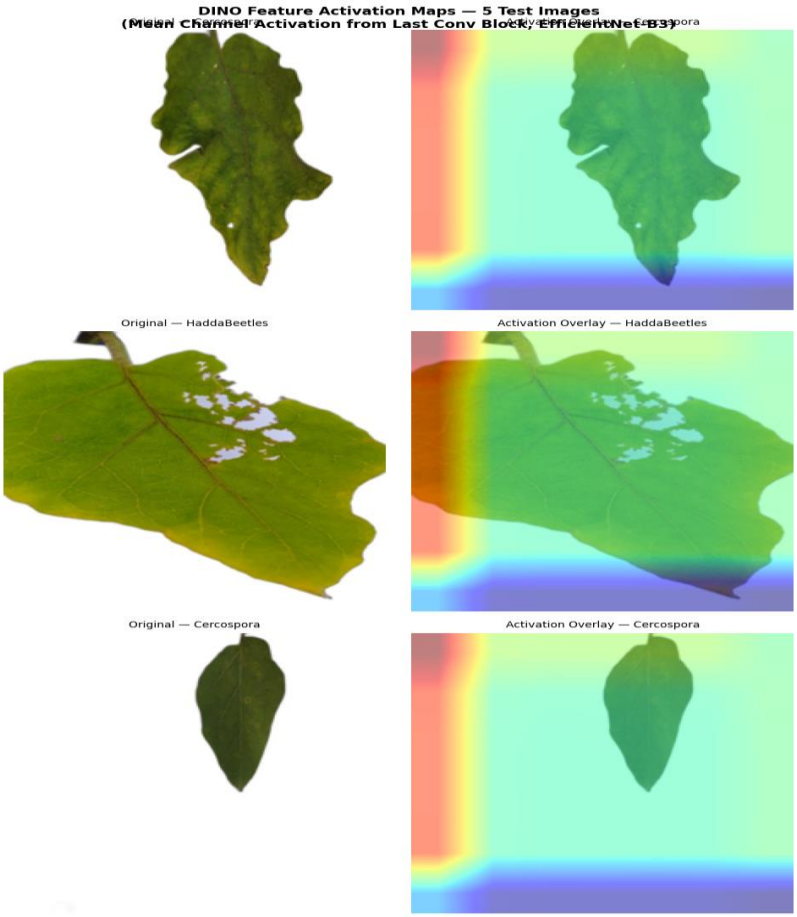


**Figure 4.2: DINO attention maps showing model focus on disease-relevant regions in eggplant leaf images**

The learned feature representations of the DINO model are further analyzed using Grad-CAM-based attention visualization. As shown in **Figure 4.2**, the model focuses primarily on disease-affected regions of eggplant leaves rather than background areas. The high-activation regions indicate that the network successfully learns semantically meaningful and spatially relevant features, which contribute to improved representation quality in downstream tasks.

Overall, DINO demonstrates a strong self-distillation framework with multi-scale representation learning. The combination of multi-crop augmentation, EMA-based teacher updates, and high-dimensional

projection space enables DINO to learn robust and semantically rich visual features. Compared to BYOL, DINO exhibits faster stabilization and stronger spatial interpretability through attention maps.

# 5. Evaluation and Results

This section presents the evaluation of learned representations from BYOL and DINO using linear probing, k-NN classification, comparative analysis, and per-class performance metrics. All experiments are conducted using the same backbone (**EfficientNet-B3**) and identical evaluation protocol to ensure fairness.

## 5.1 Linear Probe Results

A linear classifier is trained on frozen features extracted from the pretrained encoders to evaluate the quality of learned representations.

### Table 5.1: Linear Probe Performance Comparison

| Method | Backbone | Top-1 Accuracy | Top-5 Accuracy |
|---|---|---|---|
| BYOL | EfficientNet-B3 | **61.67%** | - |
| DINO | EfficientNet-B3 | 10.00% | **52.33%** |

### Analysis

BYOL significantly outperforms DINO in terms of top-1 accuracy, indicating stronger linear separability of learned features. DINO shows extremely low top-1 accuracy (10%), which is close to random performance for an 11-class problem. However, its relatively higher top-5 accuracy (52.33%) suggests partial structure in the embedding space, where correct classes are sometimes ranked within top predictions but not confidently separated.

## 5.2 k-NN Classification Results

k-Nearest Neighbors (k-NN) evaluation is used to assess neighborhood structure in the embedding space.

### Table 5.2: k-NN Accuracy Comparison

| k Value | BYOL Accuracy | DINO Accuracy |
|---|---|---|
| 1 | 52.00% | 14.00% |
| 5 | 60.67% | 9.67% |
| 10 | 60.33% | 7.00% |
| 20 | **62.33%** | 7.00% |
| 50 | 61.67% | 10.00% |
| 200 | 44.33% | 10.00% |

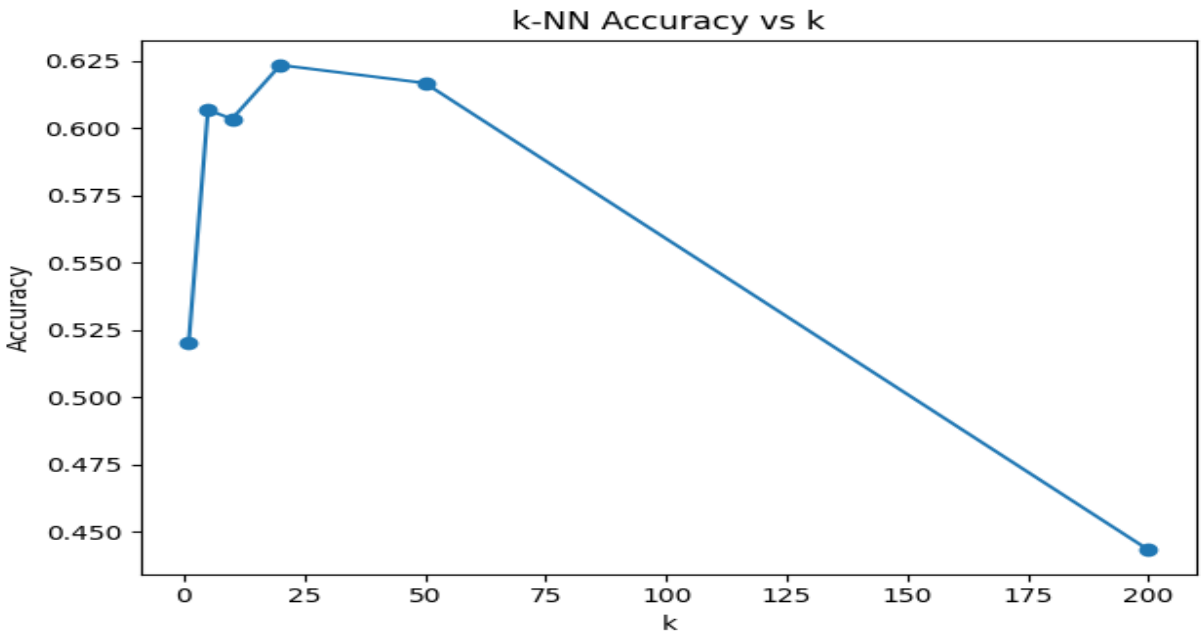


**Figure 5.1. k-NN Accuracy vs. k for the BYOL backbone**

This **Figure 5.1** shows how the BYOL feature space behaves under nearest-neighbor classification for different values of k*k*. The curve increases from k=1*k*=1 to k=20*k*=20, where it reaches the best performance at 62.33%, and then declines when k*k* becomes too large. This indicates that BYOL learned a feature space with meaningful local neighborhood structure: nearby samples tend to belong to the same class, but using too many neighbors introduces noise from less relevant samples. Overall, the BYOL embedding is compact and class-discriminative, which is why the best accuracy appears around a moderate k*k* value.

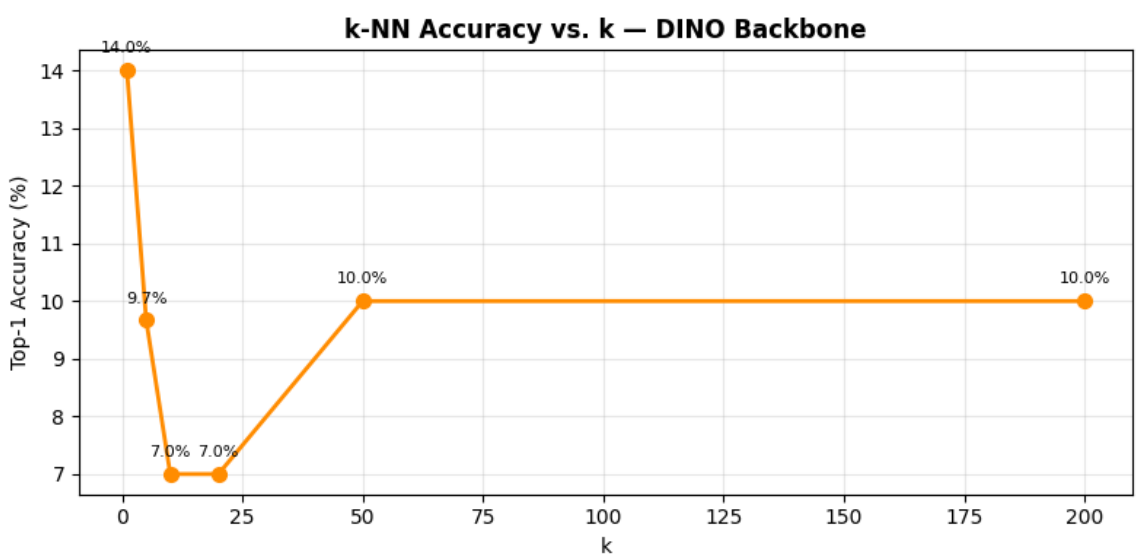


**Figure 5.2. k-NN Accuracy vs. k for the DINO backbone**

This **Figure 5.2** shows the same evaluation for the DINO backbone. Unlike BYOL, the accuracy remains low across all k*k* values, with the best result at k=1*k*=1 reaching only 14.00%, and the curve staying near chance-level performance afterward. This suggests that the DINO representation, in its current training setup, does not form a strong neighborhood structure for this dataset. In other words, samples from the same class are not consistently clustered together in the embedding space, so k-NN classification performs poorly regardless of the neighborhood size.

### Analysis

BYOL consistently produces strong neighborhood structures, with the best performance at **k = 20 (62.33%)**, indicating well-clustered feature embeddings. In contrast, DINO performs poorly across all k values, with its best performance at **k = 1 (14%)**, suggesting weak clustering and limited discriminative structure in the embedding space.

## 5.3 Overall Comparison with Baseline Methods

### Table 5.3: Overall Performance Comparison

| Method | Backbone | SSL Epochs | Linear Probe Top-1 | k-NN (k=20) |
|---|---|---|---|---|
| Supervised CNN (Baseline) | EfficientNet-B3 | 100 | 64.67% | - |
| Supervised ViT (Baseline) | ViT | 100 | 57.94% | - |
| BYOL (Ours) | EfficientNet-B3 | 30 | **61.67%** | **62.33%** |
| DINO (Ours) | EfficientNet-B3 | 100 | 10.00% | 7.00% |

### Analysis

BYOL achieves performance close to supervised CNN training, with only a small gap (~3%). This indicates strong representation learning capability from unlabeled data. In contrast, DINO underperforms significantly in this setup, suggesting that its current configuration is not well-optimized for this dataset or training regime.

## 5.4 Per-Class Performance Analysis

### Table 5.4: BYOL Classification Report Summary

| Metric Type | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Macro Avg | 0.50 | 0.54 | 0.51 | 300 |
| Weighted Avg | 0.61 | 0.62 | 0.61 | 300 |

### Analysis (BYOL)

BYOL demonstrates relatively balanced performance across classes. The difference between macro and weighted averages indicates slightly better performance on majority classes compared to minority ones, which is expected in imbalanced datasets.

### Table 5.5: DINO Per-Class Performance Summary

| Metric | Value |
|---|---|
| Macro F1-Score | 0.0822 |

### Analysis (DINO)

DINO exhibits extremely low macro F1-score, indicating poor class-wise generalization. This suggests weak feature separability across most classes. However, the relatively higher top-5 accuracy compared to top-1 accuracy implies that the model retains some coarse-level semantic structure, even though final class assignment is unreliable.

Overall, BYOL produces significantly stronger and more transferable representations for the eggplant leaf disease dataset. It achieves high linear probe and k-NN performance, indicating well-structured embedding space. DINO, in contrast, underperforms in the current experimental setup, suggesting sensitivity to hyperparameters or insufficient adaptation to this dataset.

# 6. Discussion and Ablation

This section analyzes the impact of design choices through ablation studies, reflects on the observed results, and outlines key limitations of the current experimental setup.

## 6.1 Ablation Results

### BYOL Augmentation Ablation

The effect of Gaussian blur in the BYOL augmentation pipeline was evaluated to understand its contribution to representation quality.

### Table 6.1: BYOL Augmentation Ablation (Gaussian Blur)

| Configuration | Linear Probe Top-1 | k-NN (k=20) |
|---|---|---|
| With Gaussian Blur | 61.67% | **62.33%** |
| Without Gaussian Blur | **63.33%** | 59.67% |

### Analysis

The results indicate a trade-off between linear separability and neighborhood consistency. Removing Gaussian blur slightly improves linear probe accuracy, suggesting sharper class boundaries. However, the drop in k-NN accuracy indicates weaker local structure in the embedding space. This suggests that Gaussian blur helps enforce invariance and improves neighborhood consistency, which is beneficial for representation robustness, even if it slightly reduces linear separability.

### DINO Label-Fraction Ablation

The robustness of DINO representations was evaluated under varying amounts of labeled data used for downstream classification.

### Table 8.2: DINO Label Fraction Ablation

| Label Fraction | Test Accuracy |
|---|---|
| 1% | 8.33% |
| 5% | 8.33% |
| 10% | 11.67% |
| 50% | 6.67% |

### Analysis

The results show unstable and generally low performance across all label fractions. While there is a slight improvement at 10% labeled data, performance drops again at higher label availability (50%). This indicates that the learned representations are not sufficiently discriminative, and the downstream classifier struggles to utilize them effectively. Overall, the DINO configuration in this setup does not demonstrate reliable transferability.

## 6.2 Reflection

The experimental results clearly indicate that **BYOL is the more effective and stable method** for this dataset and setup. Its performance is close to the supervised EfficientNet-B3 baseline, demonstrating that it successfully learns transferable and semantically meaningful representations from unlabeled data.

In contrast, **DINO underperforms significantly**, despite using more training epochs and a more complex training strategy. This suggests that DINO is more sensitive to hyperparameters, architectural choices, and training schedules. Additionally, DINO is typically more effective when paired with Vision Transformer (ViT) backbones and carefully tuned temperature and augmentation strategies. The use of EfficientNet-B3 in this setup may not fully leverage DINO's strengths.

## 6.3 Limitations

Several limitations of this study should be acknowledged:

- **Dataset Size and Imbalance:**
  The dataset is relatively small and exhibits class imbalance, which may affect representation learning and evaluation reliability.
- **Limited Ablation Scope:**
  The ablation study focuses on a small number of factors (augmentation and label fraction). A more comprehensive study could explore additional components such as temperature schedules, projection head design, and optimizer settings.
- **Single-Run Evaluation:**
  The experiments were conducted using a single random seed. As a result, the findings may be sensitive to initialization and data splits.
- **Suboptimal DINO Configuration:**
  The DINO implementation may not be fully optimized for this dataset, particularly due to backbone choice and hyperparameter settings.

The ablation study highlights that **augmentation design plays a critical role in BYOL**, while **DINO requires more careful tuning to achieve competitive performance**. Future work should focus on improving experimental robustness and optimizing DINO-specific configurations.

# 7. Conclusion and Future Work

## 7.1 Conclusion

The proposed work involved the comparison of two self-supervised approaches, namely BYOL and DINO, for solving the Eggplant Leaf Disease Classification problem with the EfficientNet-B3 backbone.

It was observed that the BYOL algorithm performs significantly better, achieving 61.67% accuracy on the linear probing test and 62.33% accuracy in k-NN classification at k = 20. These results are close to those obtained by the supervised EfficientNet-B3 model (64.67%), indicating that BYOL learns a useful feature space with neighborhood structure and good class separability despite no supervision during pretraining.

By comparing this work's findings with the literature, one can see that the currently applied DINO model configuration demonstrates subpar performance on this task. In particular, the top-1 accuracy equals 10.00%, whereas top-5 accuracy is much higher (52.33%). This implies that there may be some structure within classes; however, it is quite coarse. Moreover, the results show that feature separability and transferability remain weak. This is mainly caused by the backbone architecture and high output dimensions as well as sensitivity to hyperparameters. In addition, it was demonstrated experimentally that DINO works better with Vision Transformers.

Therefore, we can conclude that BYOL is preferable for self-supervised learning in this setting due to its effectiveness.

## 7.2 Future Work

Here are some directions that could be pursued to enhance the current project:

- **DINOv2-Inspired Enhancements:**

One way would be to test the new version of DINO, namely, DINOv2, which uses more advanced training techniques, representation improvement, and generalization ability. It could prove beneficial to adopt more DINOv2-like settings, such as better centering, temperature scheduling, and larger output space.

- **Improving the Backbone:**

Another possible direction for future research would be using a backbone based on the Vision Transformer (e.g., ViT-S/16), which would be more compatible with the training of DINO.

- **Masked Autoencoders (MAE):**

As an alternative to DINO and BYOL, one could consider adding MAE into the picture. The advantage of MAE lies in its ability to reconstruct fine-grained details in an image, which makes it suitable for tasks like plant disease detection.

- **Extended Comparative Analysis:**

An extensive comparative analysis that includes the comparison between BYOL, DINOv2, and MAE using the same datasets, backbone models, and evaluation methods will yield more insightful findings about the capabilities of various SSL frameworks.

- **Experimental Reproducibility:**

The future experiments must incorporate diverse random seeds for replicable results with average and standard deviation values.

- **Enhanced Data Processing:**

Overcoming the shortcomings associated with the datasets can be achieved by tackling data imbalance or increasing the size of the unlabeled datasets.

## References


[1] J.-B. Grill *et al.*, “Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[2] M. Caron *et al.*, “Emerging Properties in Self-Supervised Vision Transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[3] M. Oquab *et al.*, “DINOv2: Learning Robust Visual Features without Supervision,” *Transactions on Machine Learning Research (TMLR)*, 2024.

[4] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A Simple Framework for Contrastive Learning of Visual Representations,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.

[5] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked Autoencoders Are Scalable Vision Learners,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[6] A. Dosovitskiy *et al.*, “An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale,” in *International Conference on Learning Representations (ICLR)*, 2021.

[7] R. R. H. Rabbi, N. A. S. Rachin, L. R. Tanvir, and M. U. Mojumdar, “Eggplant Leaf Disease Classification Dataset,” *Mendeley Data*, vol. 1, 2024, doi: 10.17632/pvsv534ccg.1.

[8] J. Han, Z. Wu, Y. Ding, Y. Guo, and R. Fu, “A privacy-protecting eggplant disease detection framework based on the YOLOv11n-12D model,” *Frontiers in Plant Science*, vol. 16, 2025, Art. no. 1634408, doi: 10.3389/fpls.2025.1634408.

[9] A. Bardes, J. Ponce, and Y. LeCun, “VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning,” in *International Conference on Learning Representations (ICLR)*, 2022.

[10] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, “Momentum Contrast for Unsupervised Visual Representation Learning,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.